

Theorem (Jensen's Inequality). *Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be convex, i.e.*

$$\forall x, y \in \mathbb{R}, \forall t \in [0, 1] \quad f(tx + (1-t)y) \leq tf(x) + (1-t)f(y). \quad (1)$$

Let $x_1, \dots, x_n \in \mathbb{R}$ and non-negative weights $\lambda_1, \dots, \lambda_n$ satisfy

$$\lambda_i \geq 0 \ (i = 1, \dots, n), \quad \sum_{i=1}^n \lambda_i = 1. \quad (2)$$

Then

$$f\left(\sum_{i=1}^n \lambda_i x_i\right) \leq \sum_{i=1}^n \lambda_i f(x_i). \quad (3)$$

Equality holds iff either all the x_i are equal, or f is affine on the convex hull of $\{x_1, \dots, x_n\}$ (equivalently on the smallest interval containing the points), or the weights attached to points where f is not affine are zero.

Proof. We proceed by induction on the number of points n .

1. Degenerate cases.

- If $n = 1$, then (2) forces $\lambda_1 = 1$ and (3) reduces to $f(x_1) \leq f(x_1)$, true with equality.
- If some weight equals 1 (say $\lambda_k = 1$ and all other $\lambda_i = 0$), then $\sum_{i=1}^n \lambda_i x_i = x_k$ and $\sum_{i=1}^n \lambda_i f(x_i) = f(x_k)$; again (3) holds with equality.

Thus the statement is valid in all degenerate situations.

2. Base case $n = 2$. Write $\lambda_1 = t$, $\lambda_2 = 1 - t$ with $t \in [0, 1]$ (the condition $\lambda_1 + \lambda_2 = 1$ guarantees this). Applying the definition of convexity (1) to x_1, x_2 yields

$$f(tx_1 + (1-t)x_2) \leq tf(x_1) + (1-t)f(x_2),$$

which is exactly (3) for $n = 2$.

3. Induction hypothesis. Assume that for some integer $k \geq 2$ the inequality (3) holds for *every* collection of at most k points with non-negative weights summing to 1. We shall prove it for $n = k + 1$.

4. Inductive step. Let $x_1, \dots, x_n$ ($n = k + 1$) and weights $\lambda_1, \dots, \lambda_n$ satisfy (2). Set

$$L := \sum_{i=1}^{n-1} \lambda_i. \quad (4)$$

If $L = 0$ then $\lambda_n = 1$ and we are back to a degenerate case. Hence assume $L > 0$.

Define normalized weights for the first $n - 1$ indices:

$$\mu_i := \frac{\lambda_i}{L}, \quad i = 1, \dots, n - 1. \quad (5)$$

Clearly $\mu_i \geq 0$ and $\sum_{i=1}^{n-1} \mu_i = 1$.
Set

$$y := \sum_{i=1}^{n-1} \mu_i x_i. \quad (6)$$

By the induction hypothesis applied to the $(n-1)$ -tuple $(x_1, \dots, x_{n-1})$ with weights μ_i we have

$$f(y) \leq \sum_{i=1}^{n-1} \mu_i f(x_i). \quad (7)$$

The original convex combination can be rewritten as a two-point combination of y and x_n :

$$\begin{aligned} \sum_{i=1}^n \lambda_i x_i &= \sum_{i=1}^{n-1} \lambda_i x_i + \lambda_n x_n \\ &= L \left(\sum_{i=1}^{n-1} \mu_i x_i \right) + \lambda_n x_n = L y + \lambda_n x_n. \end{aligned} \quad (8)$$

From (2) and (4) we obtain

$$L + \lambda_n = 1. \quad (9)$$

Apply the two-point convexity (1) to the points y and x_n with weight L :

$$f(Ly + \lambda_n x_n) \leq L f(y) + \lambda_n f(x_n). \quad (10)$$

Substituting (8) for the argument of f and using (7) to bound $f(y)$ gives

$$\begin{aligned} f\left(\sum_{i=1}^n \lambda_i x_i\right) &\leq L \left(\sum_{i=1}^{n-1} \mu_i f(x_i) \right) + \lambda_n f(x_n) \\ &= \sum_{i=1}^{n-1} L \mu_i f(x_i) + \lambda_n f(x_n) \\ &= \sum_{i=1}^{n-1} \lambda_i f(x_i) + \lambda_n f(x_n), \end{aligned}$$

which is precisely (3) for n points. Hence the inequality holds for $n = k + 1$. By induction it holds for every integer $n \geq 1$.

5. Equality case. Equality in (1) occurs exactly when either the weight t equals 0 or 1 (so the convex combination reduces to a single point) or the restriction of f to the segment $[x, y]$ is affine:

$$f(tx + (1-t)y) = t f(x) + (1-t)f(y) \quad \forall t \in [0, 1].$$

In the inductive proof equality in the final inequality requires equality at the two points where we used (1) (step (10)) and where we invoked the induction hypothesis (step (7)). Consequently equality in (3) holds iff one of the following equivalent conditions is satisfied:

1. All points coincide: $x_1 = x_2 = \dots = x_n$. Then both sides of (3) equal $f(x_1)$.
2. f is affine on the convex hull of the points, i.e. there exist constants a, b such that $f(t) = at + b$ for every t in the smallest interval containing $\{x_1, \dots, x_n\}$. On such an interval the inequality (1) is an equality for any weight, and every induction step becomes an equality.
3. The weights attached to points where f is not affine are zero. Equivalently, all points with positive weight lie in an interval on which f is affine.

These are precisely the conditions stated in the theorem. $\square$

$\square$